This assignment will not be graded and is only for practice.

Key Points:

Vectors in R^2 are represented by ordered pairs (a, b).

The first element in the ordered pair denotes the X-coordinate and the second one denotes the Y-coordinate in the Cartesian plane.

The addition of two vectors $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2) \in \mathrm{R}^2$, is given by $v_1 + v_2 = (x_1 + x_2, y_1 + y_2)$.

The scalar multiplication of a vector $v = (x, y) \in \mathrm{R}^2$ with a scalar $c \in \mathrm{R}$, is given by $cv = (cx, cy)$.

1) Let $v_1 = (2, -4)$ and $v_2 = (-1, 2)$ be two vectors in R^2. Which of the following options are true? ***1 point***

[Hint: Recall that vector addition and scalar multiplication are done coordinatewise.]

- ☐ $2v_2 = (-2, 4)$
- ☐ $\frac{1}{2}v_1 = (1, -2)$
- ☐ $v_1 + 2v_2 = (0, 0)$
- ☐ $2v_1 + v_2 = (0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2v_2 = (-2, 4)$
$\frac{1}{2}v_1 = (1, -2)$
$v_1 + 2v_2 = (0, 0)$

2) Choose the set of correct options using Figure M2W1SWU1. ***1 point***

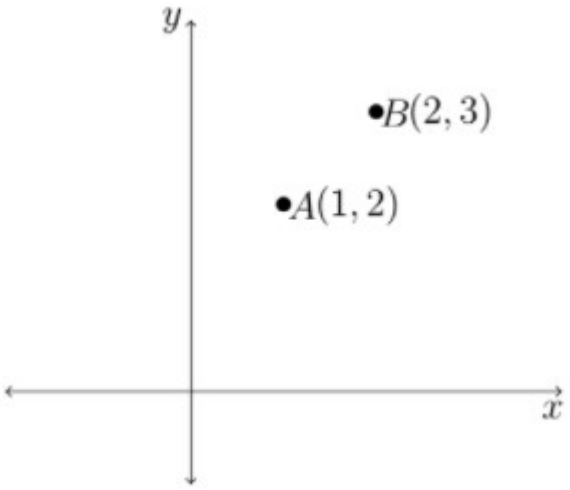


Figure M1W1SWU1

[Hint: Recall that vector addition and scalar multiplication are done coordinate-wise.]

- ☐ $2A$ is the vector $(2, 4)$.

- [ ] $3B$ is the vector $(6, 9)$.
- [ ] $A + B$ is the vector $(3, 5)$.
- [ ] $A - B$ is the vector $(-1, -1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2A$ is the vector $(2, 4)$.
$3B$ is the vector $(6, 9)$.
$A + B$ is the vector $(3, 5)$.
$A - B$ is the vector $(-1, -1)$.

2. Key points:

The addition of two vectors $V_1 = (a_1, b_1, C_1)$ and $V_2 = (a_2, b_2, C_2)$ are done coordinate-wise, as follows:

$$V_1 + V_2 = (a_1, b_1, C_1) + (a_2, b_2, C_2) = (a_1 + a_2, b_1 + b_2, C_1 + C_2)$$

The scalar multiplication of a vector $V = (a, b, c)$ by a scalar α is given by:

$$\alpha(a, b, c) = (\alpha a, \alpha b, \alpha c)$$

3) Let $V_1 = (1, 0, 0)$, $V_2 = (0, 1, 0)$ and $V_3 = (0, 0, 1)$ be three vectors and $a, b,$ and c be three real numbers (scalars). Then, which of the following is (are) true? ***1 point***

- [ ] $(a, b, c) = aV_1 + bV_2 + cV_3$
- [ ] $(a, b, c) = abV_1 + bcV_2 + caV_3$
- [ ] $(a, 0, c) = aV_1 + cV_2 + 0V_3$
- [ ] $(a, 0, c) = aV_1 + 0V_2 + cV_3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(a, b, c) = aV_1 + bV_2 + cV_3$
$(a, 0, c) = aV_1 + 0V_2 + cV_3$

3. Key Points:

Addition of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$ is given by

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{bmatrix}$$

If $R = [a \quad b \quad c]$ and $C = \begin{bmatrix} d \\ e \\ f \end{bmatrix}$, then the product RC is given by

$$[ad + be + cf]$$

Suppose $A = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$ and $B = [C_1 \quad C_2 \quad C_3]$, where R_i's denote the rows of matrix A and C_i's denote the columns of matrix B. Moreover, assume that the number of columns of A and the number of rows of B are the same. The product AB is given by,

$$AB = \begin{bmatrix} R_1C_1 & R_1C_2 & R_1C_3 \\ R_2C_1 & R_2C_2 & R_2C_3 \end{bmatrix}$$

Addition of two matrices A and B is defined if both A and B have the same number of rows and the same number of columns. If both the matrices A and B have m rows and n columns, then the matrix $A + B$ also has m rows and n columns

If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then AB is well-defined and is an $m \times p$ matrix.

4) Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}$, $Q = [1 \quad 4 \quad -9]$, $R = [0 \quad -3 \quad 10]$, $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$ ***1 point***

[Hint: If A is a matrix of order $m \times n$ and B is a matrix of order $n \times p$, then the order of AB is $m \times p$.]

- ☐ The matrix PD is of order 3×1.
- ☐ The matrix PD is of order 1×3.
- ☐ The matrix QD is of order 3×3.
- ☐ The matrix QD is of order 1×1.
- ☐ The matrix DQ is of order 3×3.
- ☐ The matrix DQ is of order 1×1.
- ☐ The product QD is not defined.
- ☐ The product QR is not defined.
- ☐ The addition $P + Q$ is not defined.
- ☐ The addition $P + D$ is not defined.

No, the answer is incorrect.

Score: 0

Feedback:

• P is a 3 × 3 matrix, D is a 3 × 1 matrix. Think about the order of P D.
• Q is a 1 × 3 matrix, D is a 3 × 1 matrix. Think about the order of QD and DQ.
• Q is a 1 × 3 matrix, R is a 1 × 3 matrix. Think whether is it possible to define QR or not.
• Figure out which pair of matrices has the same number of rows as well as the same number of columns.

Accepted Answers:

The matrix PD is of order 3×1.
The matrix QD is of order 1×1.
The matrix DQ is of order 3×3.
The product QR is not defined.
The addition $P + Q$ is not defined.
The addition $P + D$ is not defined.

5) Suppose $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 5 \\ 3 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2 & 3 \\ 2 & 5 & 4 \end{bmatrix}$. Which of the following options are correct? **1 point**

[Hint: Multiplication of matrices $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$ is given by $AB =$
$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{bmatrix}$
Similarly, BA can be calculated.]

- [ ] $AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 26 & 29 \\ 11 & 29 & 34 \end{bmatrix}$
- [ ] $AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \\ 11 & 26 & 34 \end{bmatrix}$
- [ ] $BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \\ 14 & 22 & 45 \end{bmatrix}$
- [ ] $BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 22 & 13 \\ 14 & 13 & 45 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Feedback:

Observe that B = A^T (where A^T denotes the transpose of A).

Accepted Answers:

$AB = \begin{bmatrix} 5 & 10 & 11 \\ 10 & 29 & 26 \\ 11 & 26 & 34 \end{bmatrix}$

$BA = \begin{bmatrix} 10 & 9 & 14 \\ 9 & 13 & 22 \\ 14 & 22 & 45 \end{bmatrix}$

6) Suppose $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ Which of the following options are true? ***1 point***

[Hint: Calculate A^2 and B^2]

- ☐ $A^2 = I$
- ☐ $A^2 = A$
- ☐ $B^2 = I$
- ☐ $B^2 = 0$

No, the answer is incorrect.

Score: 0

Feedback:

Correctly solving this question will give a hint to solve Week 4 Lecture 1.2 : Activity Question 6.

Accepted Answers:

$A^2 = I$
$B^2 = 0$

4. Key points:

Consider a system of linear equations as follows:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$,

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

A is called *Coefficient matrix*.

Suppose there are m number of equations and n number of variables in the system of linear equations, then the coefficient matrix will be an $m \times n$ matrix, x will be an $n \times 1$ and b will be an $m \times$

1 matrix.

7) Consider a system of linear equations (System 1): **1 point**

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$

If the matrix representation of system (1) is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

[Hint: If a variable is absent in an equation then the coefficient of the variable must be taken as 0.]

- ☐ $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$
- ☐ $b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$
- ☐ $A = \begin{bmatrix} -2 & -1 & 0 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$
- ☐ $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

8) Consider a system of equations: **1 point**

$$2x_1 + 3x_2 = 6$$
$$-2x_1 + kx_2 = d$$
$$4x_1 + 6x_2 = 12$$

Choose the set of correct options.

[Hint: Observe that third equation is a multiple of the first one (Dividing by 2 from both the sides of the third equation gives the first equation). So it is enough to check the solutions for the first and second equation.]

- [ ] $Ax = b$ represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$, and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$
- [ ] The system has no solution if $k = -3, d = 0$.
- [ ] The system has a unique solution if $k = 3, d = 0$.
- [ ] The system has infinitely many solutions if $k = -3, d = 6$.
- [ ] The system has infinitely many solutions if $k = -3, d = -6$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$Ax = b$ represents the above system, where $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$, and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$

The system has no solution if $k = -3, d = 0$.
The system has a unique solution if $k = 3, d = 0$.
The system has infinitely many solutions if $k = -3, d = -6$.

5. Key points:

Let A be a 2×2 matrix as follows:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$det(A) = ad - bc$$

Let A be a 3×3 matrix as follows:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$det(A) = a_{11} \times det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) - a_{12} \times det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}\right) + a_{13} \times det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\right)$$

i.e., $det(A) = a_{11} \times (a_{22}a_{33} - a_{32}a_{23}) - a_{12} \times (a_{21}a_{33} - a_{23}a_{31}) + a_{13} \times (a_{21}a_{32} - a_{22}a_{31})$

(Expanding with respect to first row)

9) Find the determinant of the matrix $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$

[Hint: Use the formula for finding the determinant of a matrix of order 2×2.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

10) Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ be a 3×3 matrix. Which of the following is(are) correct? ***1 point***

[Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

- ☐ $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$
- ☐ $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$
- ☐ $det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$
- ☐ $det(A) = 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) - 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$

$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$

$det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$

6. Key points:

Row operations and relation with determinants:

- Type 1: Interchanging two rows of a matrix changes the sign of the determinant.
- Type 2: Multiplying a real number with a row and adding it to some other row does not change the determinant.
- Type 3: The determinant of a new square matrix obtained by multiplying a real number c with a row of square matrix of order n, is c times the determinant of the original matrix.

Same types of column operations, as discussed above for row operations, gives same type of relation with the determinants.

If two rows or two columns of a matrix are equal, then the determinant of the matrix is 0.

If all the elements of any row or any column of a matrix are 0, then the determinant of the matrix is 0.

$det(A) = det(A^T)$, where A^T denotes the transpose of A.

$det(AB) = det(BA) = det(A)det(B)$.

11) Let A be a 3×3 matrix with non-zero determinant. If $det(2A) = k\ det(A)$, then what will be the value of k? ***1 point***

[Hint: If a scalar (c) is multiplied with one row of a matrix A, then the determinant of the new matrix will be c times the determinant of A.]

- ○ 2
- ○ 4
- ○ 8
- ○ 16

No, the answer is incorrect.

Score: 0

Accepted Answers:

8

12) Let A be a 2×2 matrix, which is given as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. Define the following matrices : ***1 point***

$$B = \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}, C = \begin{bmatrix} a_{11} - a_{12} & a_{12} \\ a_{21} - a_{22} & a_{22} \end{bmatrix},$$

$$D = \begin{bmatrix} a_{11} + a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}, E = \begin{bmatrix} a_{11} - a_{21} & a_{12} + a_{22} \\ a_{21} & a_{22} \end{bmatrix}$$

Which of the matrices among $B, C, D,$ and E have the same determinant as that of the matrix A, for any real numbers $a_{11}, a_{12}, a_{21}, a_{22}$?

[Hint: Try to observe whether B, C, D, and E can be obtained from A by row or column operations of Type 2 mentioned in the key points.]

- ○ B and D
- ○ B and E
- ○ B and C
- ○ D and E
- ○ C and E

No, the answer is incorrect.

Score: 0

Feedback:

• B can be obtained by a row operation of Type 2.
• C can be obtained by column operation of Type 2.
• D cannot be obtained by a row operation or column operation of Type 2. Check the determinant of D and see whether it matches with determinant of A or not.
• E cannot be obtained by a row operation or column operation of Type 2. Check the determinant of E and see whether it matches with determinant of A or not.

Accepted Answers:

B and C

13) If all the elements of a 3×3 real matrix A are the same, then which of the following is (are) correct? ***1 point***

[Hint:
· If two rows or two columns of a matrix are equal, then the determinant of the matrix is 0.
· $det(A) = det(A^T)$, where A^T denotes the transpose of A].

- [ ] Determinant of matrix A is 0.
- [ ] Determinant of matrix A cannot be determined from the given information.
- [ ] Determinant of matrix A will be the sum of the elements of a row.
- [ ] Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.
- [ ] Determinant of matrix $A + A^T$ cannot be determined from the given information, where A^T denotes the transpose of A.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Determinant of matrix A is 0.
Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.

14) Let $A, B,$ and C be 3×3 real matrices. Which of the following options is (are) correct? ***1 point***

[Hint: · $det(A) = det(A^T)$, where A^T denotes the transpose of A.
· $det(AB) = det(BA)$.]

- [ ] $det(ABC) = det(A)det(B)det(C)$
- [ ] $det(A^3) = det(A)^3$
- [ ] $det(A + B + C) = det(A) + det(B) + det(C)$
- [ ] $det(AB^T) = det(A)\ det(B)$, where B^T denotes the transpose of B.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(ABC) = det(A)det(B)det(C)$
$det(A^3) = det(A)^3$
$det(AB^T) = det(A)\ det(B)$, where B^T denotes the transpose of B.

Your score is: 0/14